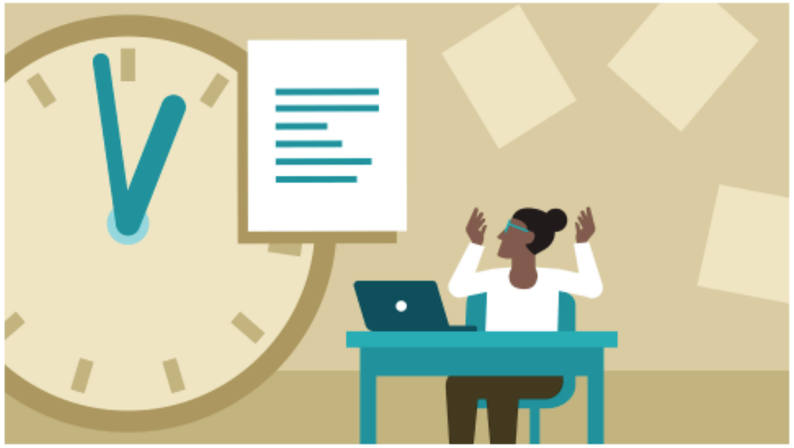
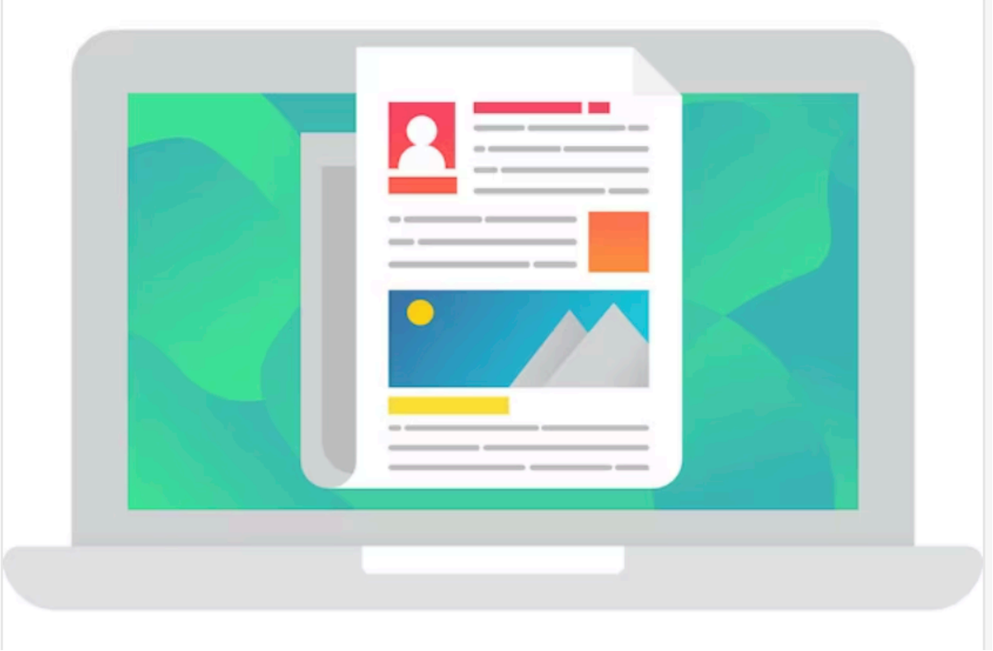
Assignment 1 Specification

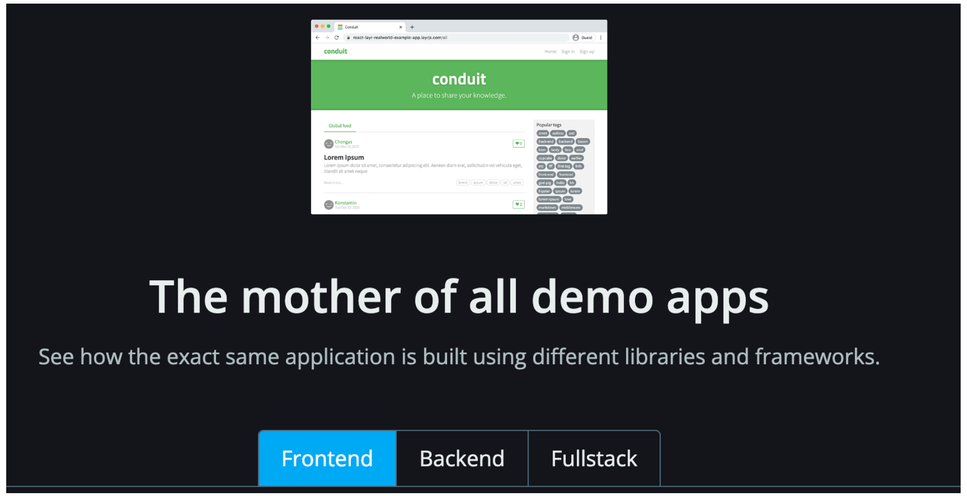


Concept, detail and grading spectrum for Assignment 1

Assignment Motivation & Philosophy

- Assignments present multiple opportunities:
  - Crystallise knowledge on relevant module(s)
  - Opportunity for creativity & innovation
  - Potentially important portfolio contribution
- Its not over when submitted, presented and graded
  - Ongoing maintenance & enhancements
  - Periodically rebuild using new technology stack





https://github.com/gothinkster/realworld
https://codebase.show/projects/realworld

Assignment 1 Topics

01: Node

The fundamentals of Node.
A first Node application.

02: Hapi.js

The Hapi application
Framework

03: Joi

Validation with Joi

04: TDD

Test Driven Development

05: Models

Incorporate Mongo Models

06: Style Guide

The AirBnB style guide

07: APIs

Expose an API from an
application

08: REST

Resful API implementations

09: OpenAPI

Documentating
OpenAPI/Swagger APIs

10: JWT

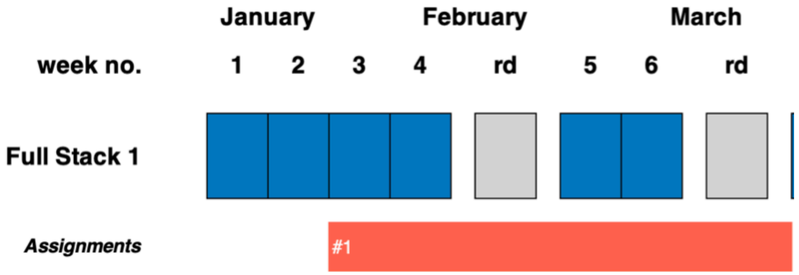
Authenticating APIs with
JSON Web Tokens

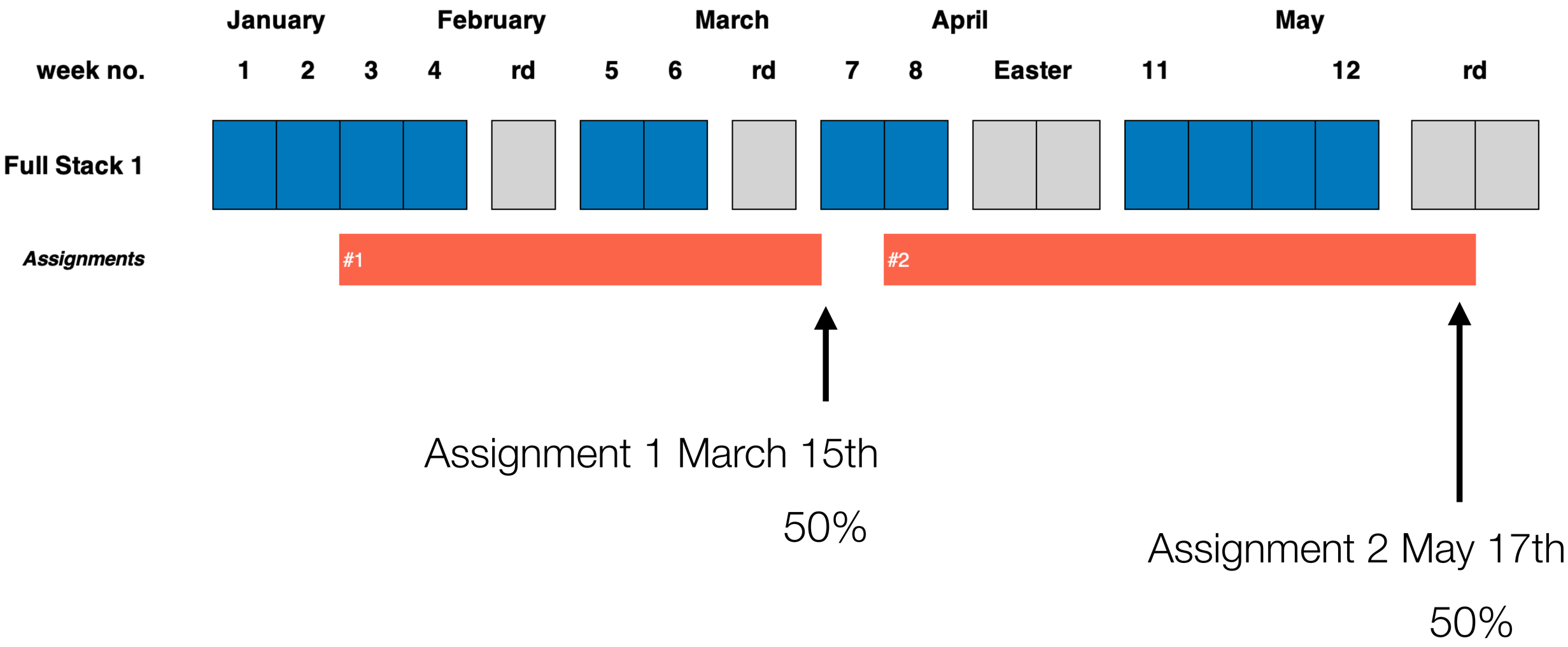
11: Seeding

Seeding the Database

12: Deployment

Deploying to Glitch, Render
& Cloud Atlas




January
February
March
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May
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1
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Easter
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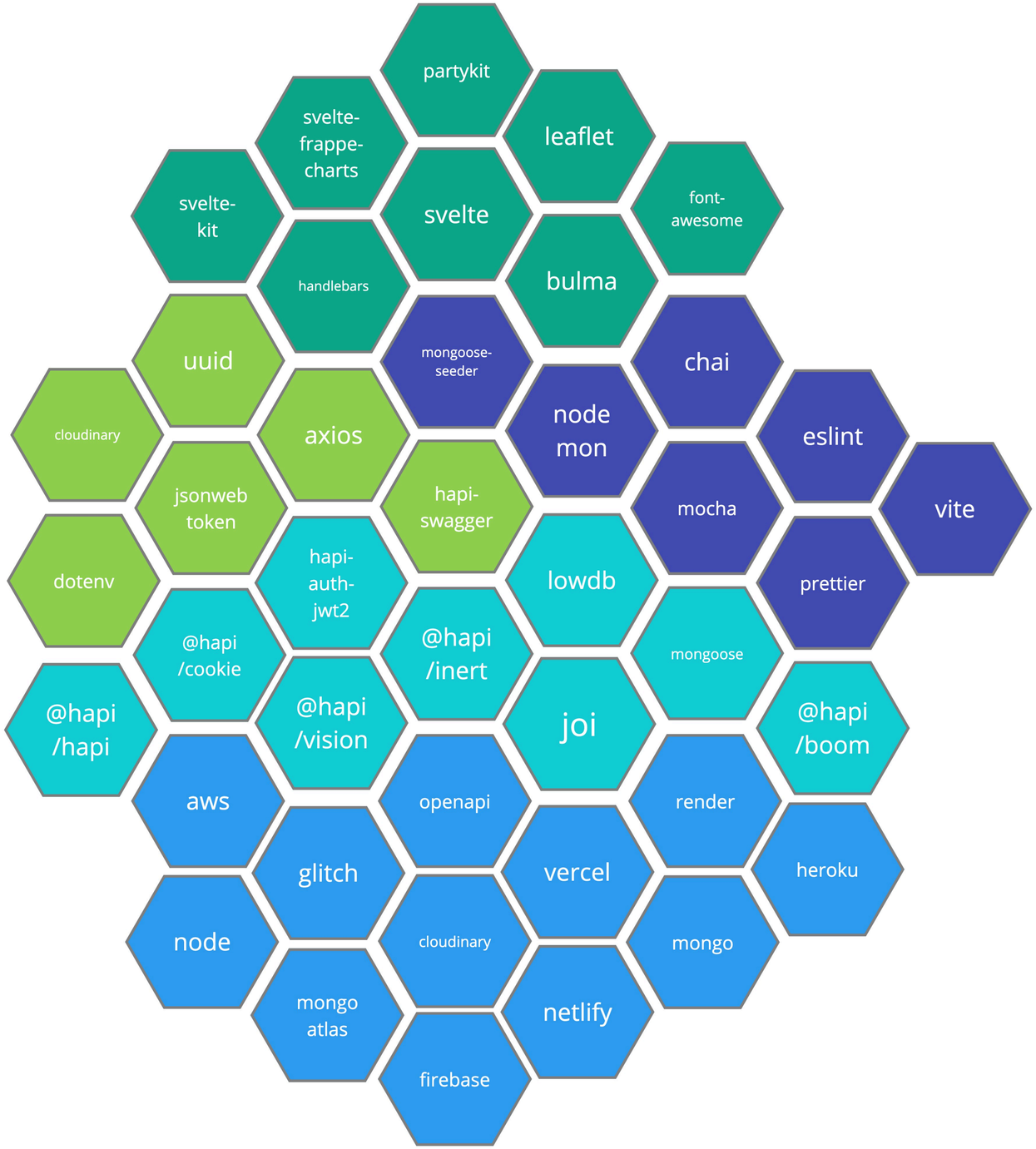
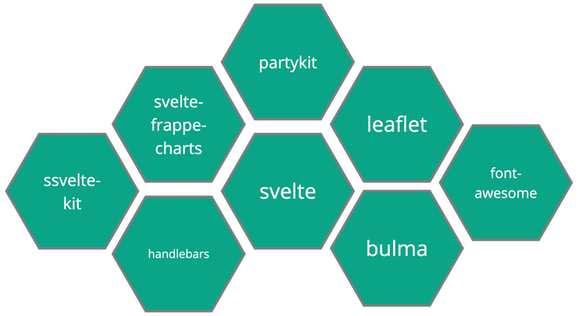
Full Stack 1
Assignments
#1
#2
Assignment 1 March 15th
50%
Assignment 2 May 17th
50%

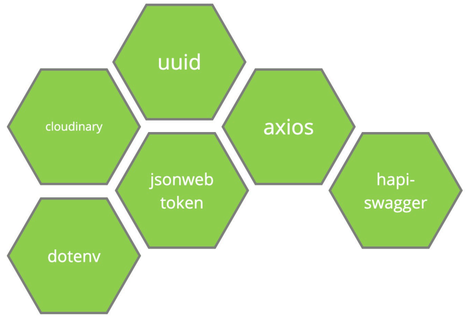
Full Stack Components & Services



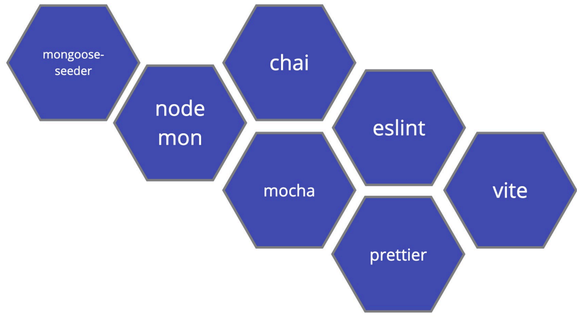
Front End



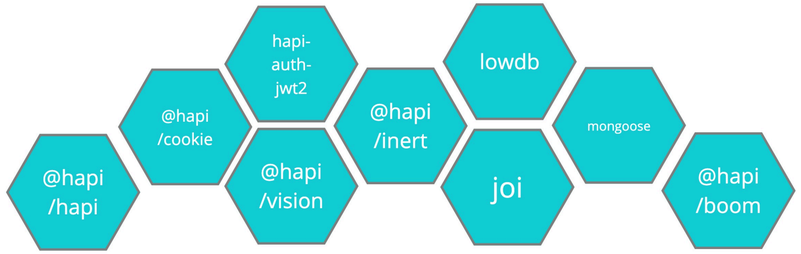
Components



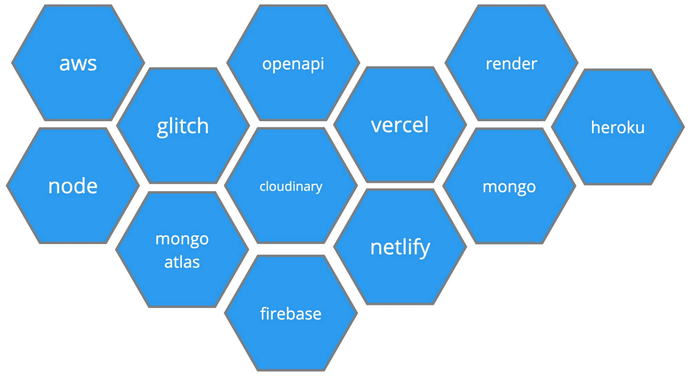
Tools



Back End



Infrastructure



PlaceMark

A point of interest, or POI, is a specific point location that someone may find useful or interesting.

"A "point of interest" (POI) is a location for which information is available. A POI can be as simple as a set of coordinates, a name, and a unique identifier, or more complex such as a three-dimensional model of a building with names in multiple languages, information about opening and closing hours, and a civic address. POI data has many applications, including augmented reality browsers, location-based social networking games, geocaching, mapping and navigation systems."

http://www.opengeospatial.org/projects/groups/poiswg

https://en.wikipedia.org/wiki/Point_of_interest

https://en.wikipedia.org/wiki/Point_of_interest

PlaceMark Examples

- Landscape feature
- National monument
- Walking Trail
- Bridge
- Tree
- Venue
- Ringfort
- Dolmen
- River
- Bog
- Island
- Forest
- Beach
- Museum
- Climb
- Cycleway
- Birdwatching Locations
- Diving

Product Feature Sets

POI Sharing / Community Features

Community

Richer Media Experience + User Management

Enhanced

Essential Data Structure & Support Service

Core

- You are free to innovate an the type, range and variety of features
- These categories are to stimulate your brainstorming
- Some ideas on the next slides…

Information Model

- Placemark
  - Name
  - Category
  - Description
  - Analytics
- User
  - Basic user details

Features

- Sign up / log in to service
- Create, Read, Update, Delete
- Query
- Group into Categories

Core

Information Model

- Placemark
  - Images
  - Location
  - Weather
- User
  - Types

Features

- Placemark images
- Placemark Map Location
- Admin
  - Admin user
  - Admin dashboard

Enhanced

Information Model

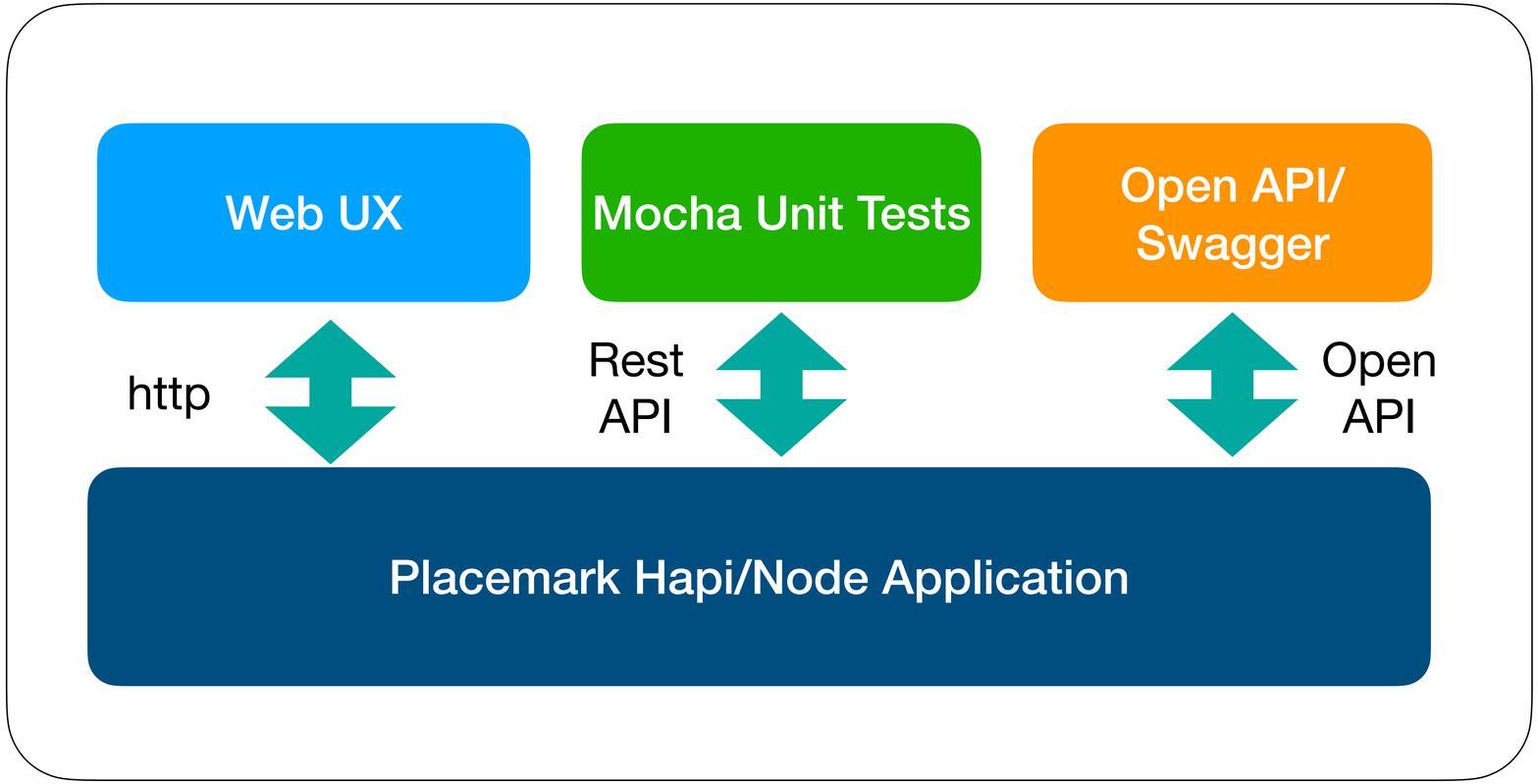
- Placemark
  - Creator / Editor
  - Rating
  - Reviews
  - Discussion
- User
  - Interaction information

Features

- Share
- Review & Rate
- Notice Board
- Live Updates
- Analytics

Community

PlaceMark 1.0 Architecture
(Assignment 1)



Placemark 1.0 Feature Set

Enhanced

Core

Implement POI Core + selected features of POI Enhanced

Support the following User Management Features:

- Sign up / Login in / Delete Account
- User Admin Dashboard with simple analytics

Support the following POI Characteristics:

- Name, Description, Category, Location, Image

Support the following app features:

- Create, Read, Update & Delete POIs
- Organise PlaceMarks into categories

API

- Endpoints + tests
- Swagger Documentation
- JWT Security

	Accounts	*Placemark Features*	*API/Tests*	*Models*	*Deployment*	*Git*
Level 1	Signup / Login	Name	Basic API	Mem	Locahost	git + commit History
Level 2	Cookie Authentication/ User Settings	Description, Location (lat,lng)	Core unit Tests	JSON	Render	Structured readme
Level 3	Basic Admin Account List/Remove Users	Categories	Open API (Swagger) + tests	Mongo	Cloud Atlas	Tagged releases
Level 4	Admin Dashboard + analytics	Images	JWT + tests	Firebase	Amazon	Git Flow
Level 5	Propose 2 additional capabilities from the above categories					

Getting Started

- Identify a set of suitable Point of Interest concept for your app. Consider how they would be created, grouped, shared, Consider the value of the service you are inventing. To whom would it be of interest? What would they do with it?

- Once you have determined the Poi concept, compose a small set of example data you will use for testing

- Sketch out a web interface - using pencil & paper of a perhaps even a simple wireframe too like Excallidraw

- Revisit the Playtime labs - perhaps consider using the latest completed lab as a starter project.

- The User model can probably be adopted as is, however you will need to implement a new model to match your PoI concept.

- Get the user system working + some UX for creating and viewing your PoI.

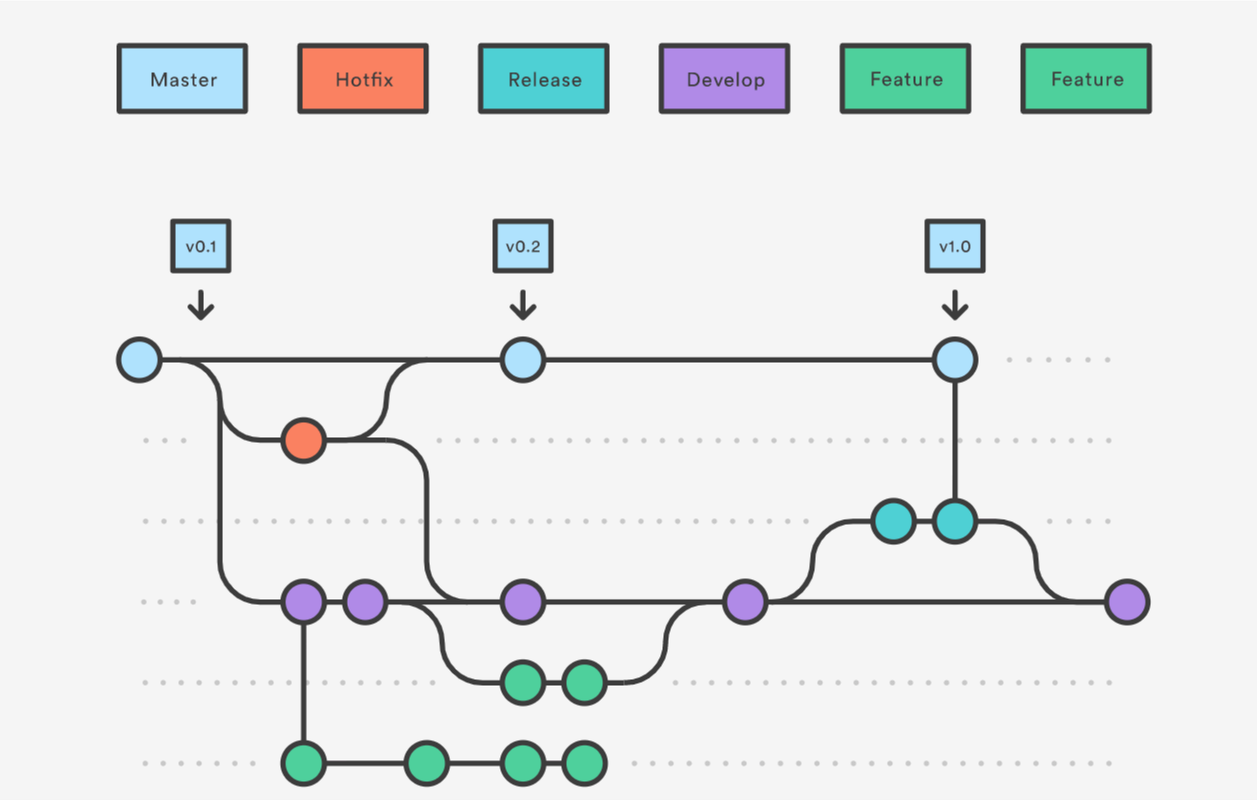
- Grouping / Categorising PoIs: What would a group represent? What would the UX for this look like? Prototype an approach.

- At this stage, you might have a sufficient set of ideas/concepts/tasks in play to consider organising your work in some way. This could be a simple whiteboard - or you might like something more structured. Github Projects is worth evaluating. It can be used without necessarily using advanced git features

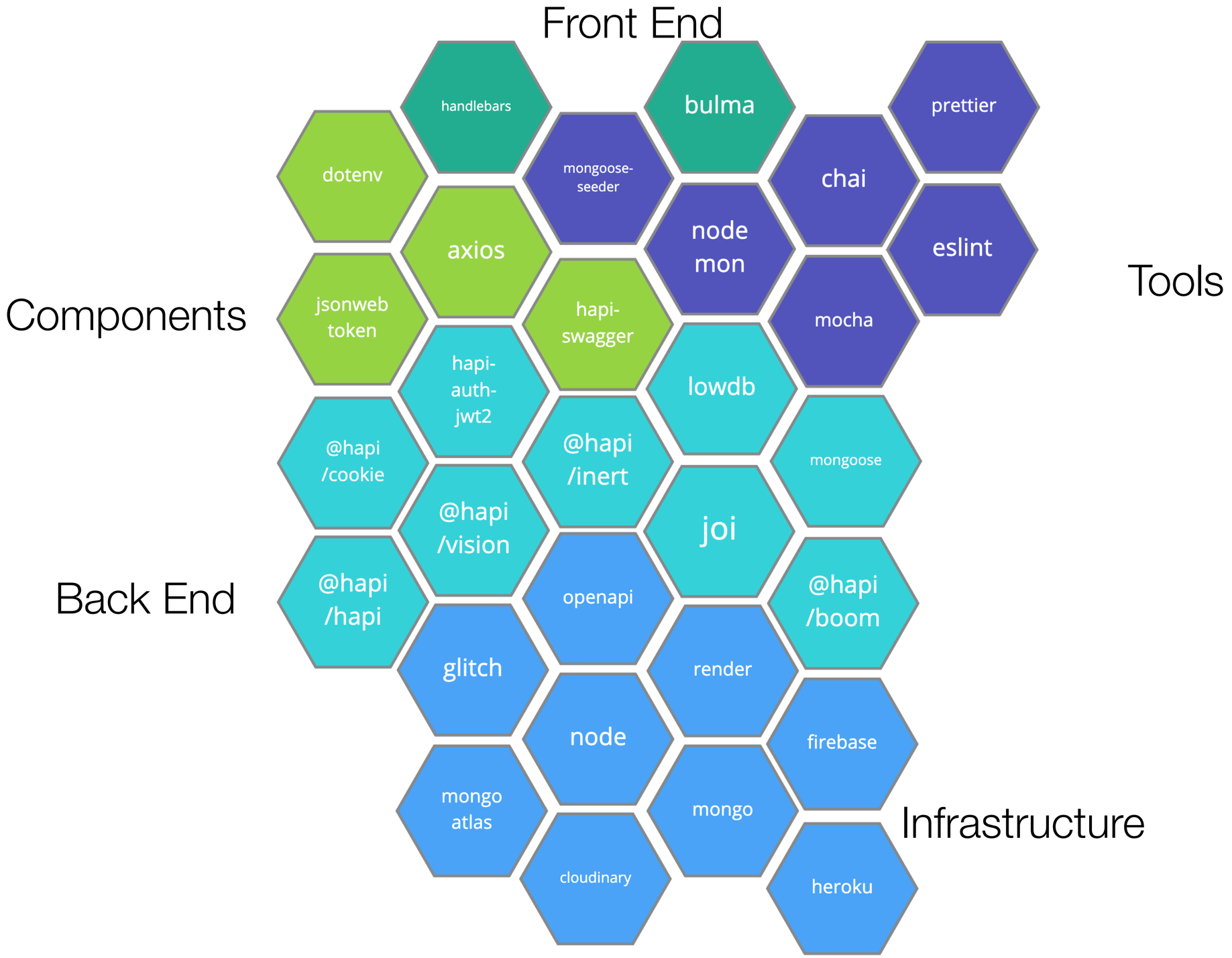
Gitflow Workflow

Gitflow Workflow is a Git workflow design that was first published and made popular by Vincent Driessen at nvie. The Gitflow Workflow defines a strict branching model designed around the project release. This provides a robust framework for managing larger projects.

Gitflow is ideally suited for projects that have a scheduled release cycle. This workflow doesn't add any new concepts or commands beyond what's required for the Feature Branch Workflow. Instead, it assigns very specific roles to different branches and defines how and when they should interact. In addition to `feature` branches, it uses individual branches for preparing, maintaining, and recording releases. Of course, you also get to leverage all the benefits of the Feature Branch Workflow: pull requests, isolated experiments, and more efficient collaboration.



https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow


Front End
handlebars
bulma
prettier
dotenv
mongoose-
seeder
chai
axios
node
mon
eslint
Tools
Components
jsonweb
token
hapi-
swagger
mocha
hapi-
auth-
jwt2
lowdb
@hapi
/cookie
@hapi
/inert
mongoose
@hapi
/vision
joi
Back End
@hapi
/hapi
openapi
@hapi
/boom
glitch
render
node
firebase
mongo
atlas
mongo
Infrastructure
cloudinary
heroku

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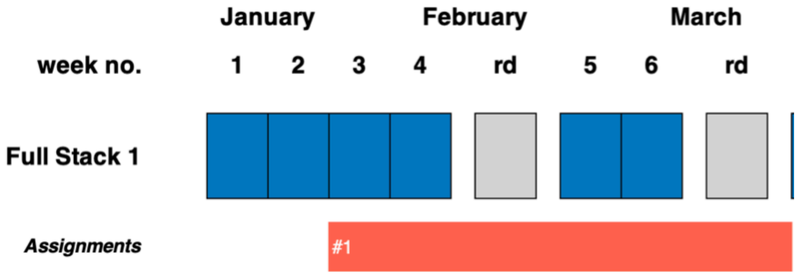
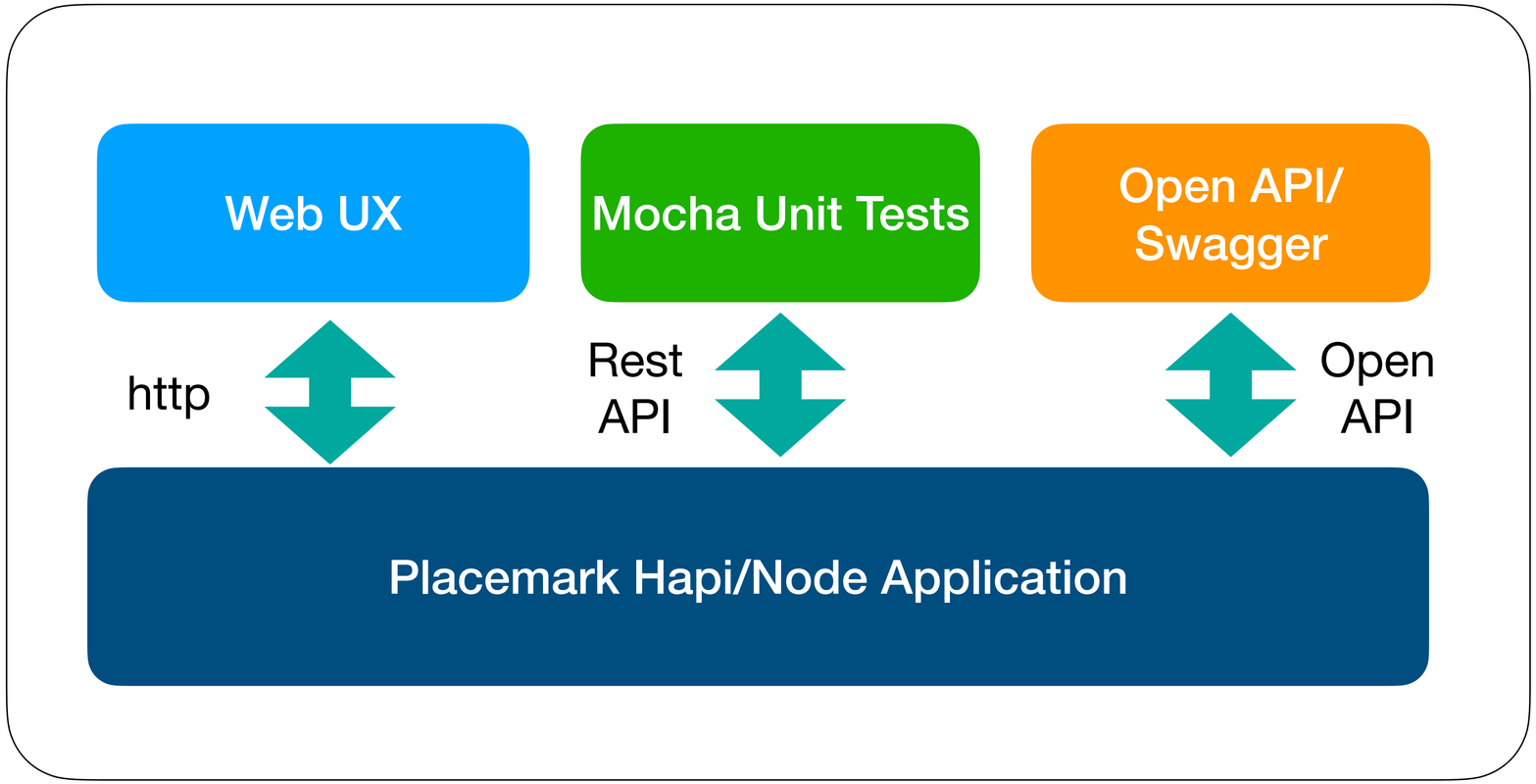
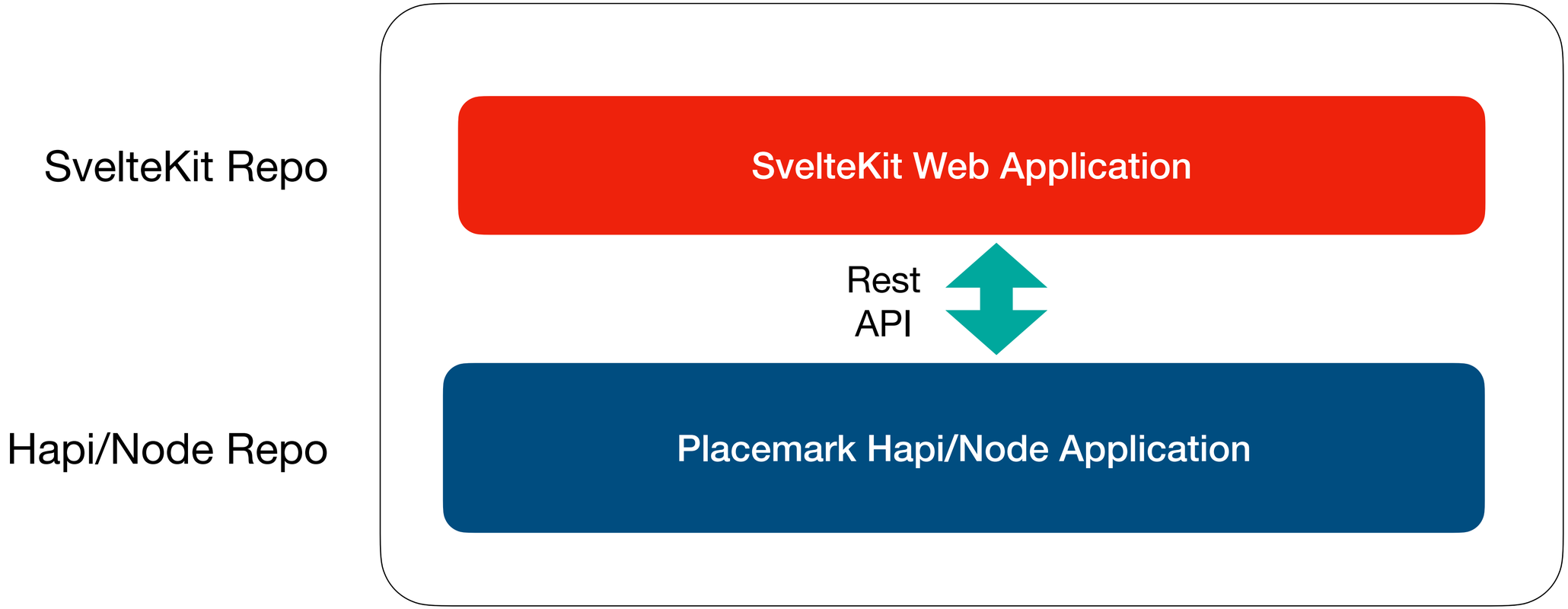


PlaceMark 1.0 Architecture
(Assignment 1)

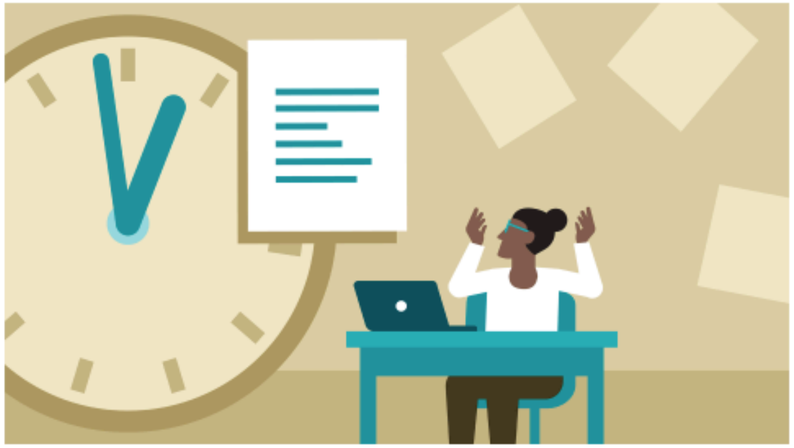
Submitted as a single repo



PlaceMark 2.0 Architecture
(Assignment 2)



Assignment 1 Specification



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